Lana Nakai

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Education

University of British Columbia — Vancouver, BC
Bachelor of Science in Computer Science (+ Co-op) | GPA 92.1% — *Sep. 2025 – May 2030 (expected)*

Technical Skills

Languages: C++, Python, Java, HTML, CSS, JavaScript, R, Racket, C
Frameworks: ROS2, PX4, gtest, pytest, JUnit, Tailwind CSS
Developer Tools: Git, Docker, Secure Shell (SSH), React.js, Vite, CMake, Bazel, Maven, VS Code, Cursor, Eclipse, Claude Code
Relevant Courses: Software Construction, Discrete Mathematics, Data Structures and Algorithms (C++), Intro to Computer Systems, Physics (Mechanics), Multivariable Calculus, Matrix Algebra, Intro to Data Science, Intro to Statistics

Technical Experience

Software Developer — Sep. 2026 – Present
UBC UAS | C++, Python, ROS, Docker, Git — *Vancouver, BC, Canada*

- Collaborating with about 20 software engineers to build competitive end to end drone autonomy
- Using ROS to integrate software components including computer vision, ML, and telemetry
- Utilize WebRTC to integrate video streaming pipeline with ground communications

Software Engineering Intern — Jul. 2026 – Sep. 2026
CHAOS Industries | C++, Python, Docker, Git — *San Francisco, CA, USA*

- Tested custom robotics framework built on Rust and C++ by applying it to a real use case (detailed in Atropos section)
- Analyzed framework time and memory efficiency using **bash scripting**
- Communicated with mentors frequently while debugging in the company monorepo, used Git best practices

Software Engineering Intern — Jun. 2026 – Jul. 2026
Atropos Group | C++, Python, ROS2, SSH, Docker, Git — *Sunnyvale, CA, USA*

- Developed, tested (HIL/SIL), and deployed data logging software from a Ground Control station to autopilot computer
- Built telemetry streaming over UDP for **real-time** flight data visualization
- Optimized logic so **+60,000 messages** could be analyzed almost instantaneously for verification
- Acquired by Chaos — personally contributed to smooth integration into new company

Projects

Activity Generator | *Java, JUnit, JSON, React.js, Next.js, TypeScript, Spring boot, Maven* — Feb 2026 - Present

- Designed and built an activity generating card game with JSON data serialization, following OOP principles
- Converted Java into a **REST API** using Spring Boot, implemented data persistence using JPA to store data in a H2 database
- Originally used Java Swing and CSS for GUI, converted frontend to be built on TypeScript and React

Personal Website | *React.js, Tailwind CSS v4, JavaScript, Vite, Vercel* — May. 2026

- Built a personal website to showcase my projects, deployed with Vercel
- Implemented EmailJS API for my personal contact card, keeping secret information private using a .env file

Motion | *HTML, CSS, JavaScript, Git* — Apr. 2026

- **First place** project in the youCode 2026 hackathon's beginner stream, built to be accessible and tailored for non-profits
- **Full stack developer**: built the front-end using HTML, CSS, and Javascript, and implemented a back-end for user log-in using Supabase.

Community Involvement

CPSC 110 Undergraduate Teaching Assistant — Jan. 2026 – Present
University of British Columbia — *Vancouver, BC*

- Teach and support students, focusing on software development through recursive theory and systematic design
- Work in a team to **lead** labs of about 30 students, communicating how to implement algorithms and graphs in functional programming and why it works.
- Give lectures to roughly 60 students weekly based on their needs.